

谭钦翰

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教育经历

普林斯顿大学	2020-2025
博士-电子与计算机工程	导师: Sharad Malik
普林斯顿大学	2020-2022
硕士-电子与计算机工程	导师: Sharad Malik
浙江大学	2016-2020
学士-计算机科学与技术	导师: 卜凯

工作经历

苹果	2025/9-今
形式化验证工程师	
英伟达	2024/6-2024/8
形式化验证实习生	

研究方向

- 形式化验证算法
- 芯片安全验证
- 芯片侧信道攻击与防御

论文 (*代表共同第一作者)

Tingzhen Dong*, Kenpeng Wang*, Yuheng Yang, Yu-Wei Fan, **Qinhan Tan**, Thomas Bourgeat, Sharad Malik, Mengjia Yan. "The Interplay of Effective Model Checking and Secure Processor", *47th IEEE Symposium on Security and Privacy, (SP 2026)*

Yuheng Yang*, **Qinhan Tan***, Thomas Bourgeat, Sharad Malik, Mengjia Yan. Compass: "Navigating the Design Space of Taint Schemes for RTL Security Verification", *31th ACM International Conference on Architectural Support for Programming Languages and Operating Systems(ASPLOS 2026)*

Qinhan Tan, Akash Gaonkar, Yu-Wei Fan, Aarti Gupta, Sharad Malik. SecIC3: Customizing IC3 for Hardware Security Verification *2026 Design, Automation Test in Europe Conference (DATE 2026)* (最佳论文候选)

Qinhan Tan*, Yuheng Yang*, Thomas Bourget, Sharad Malik, Mengjia Yan, "RTL Verification for Secure Speculation Using Contract Shadow Logic", *30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS 2025)*

Jinyan Xu, Yangye Zhou, Xingzhi Zhang, Yinshuai Li, **Qinhan Tan**, Yinqian Zhang, Yajin Zhou, Rui Chang, Wenbo Shen. DejaVuzz: Disclosing Transient Execution Bugs with Dynamic Swappable Memory and Differential Information Flow Tracking Assisted Processor Fuzzing. *30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS 2025)*

Qinhan Tan*, Yonathan Fisseha*, Shibo Chen*, Lauren Biernacki, Jean-Baptiste Jeannin, Sharad Malik, Todd Austin, "Security Verification of Low-Trust Architectures", *30th ACM Conference on Computer and Communications Security (CCS 2023)* (杰出论文奖)

Qinhan Tan, Aarti Gupta, Sharad Malik, "Usage-Based RTL Subsetting for Hardware Accelerators", *41th International Conference on Computer-Aided Design (ICCAD 2022)*

Qinhan Tan*, Zhihua Zeng*, Kai Bu and Kui Ren, "PhantomCache: Obfuscating Cache Conflicts with Localized Randomization", *27th Network and Distributed System Security Symposium (NDSS 2020)*

Qinhan Tan, Seetal Potluri, Aydin Aysu, "Efficacy of Satisfiability-Based Attacks in the Presence of Circuit Reverse-Engineering Errors", *IEEE International Symposium on Circuits and Systems (ISCAS 2020)*

专利

中国发明专利

- 1. 卜凯, 谭钦翰, 曾治华, "一种基于局部随机映射的缓存侧信道攻击防御方法 [发明专利]", 专利号 CN201910666998.1.

所获荣誉

CCS 杰出论文奖	2023
普林斯顿大学工程与应用科学学院奖学金	2023
普林斯顿大学电子与计算机科学系优秀助教奖	2023
浙江省政府奖学金	2019
浙江大学一等学业奖学金	2019